

Test Report for Dhyana 2100

1. Test Environment

Window10

CPU: i9-10900X

RAM: 64GB

Raid Array: High point SSD7505 PCIE Gen4 + 990PRO*4

Written speed in Benchmark: 6982MB/s



Date:2025.10.16

2. Capture Setup

We regenerated the phenomenon in our camera on live mode. (We normally recommend our customer to use trigger for capturing image. The frame drop might be caused due to CPU source distribution uneven in live mode.)

Total Frame: 150000

Image Stack: 1 or 10

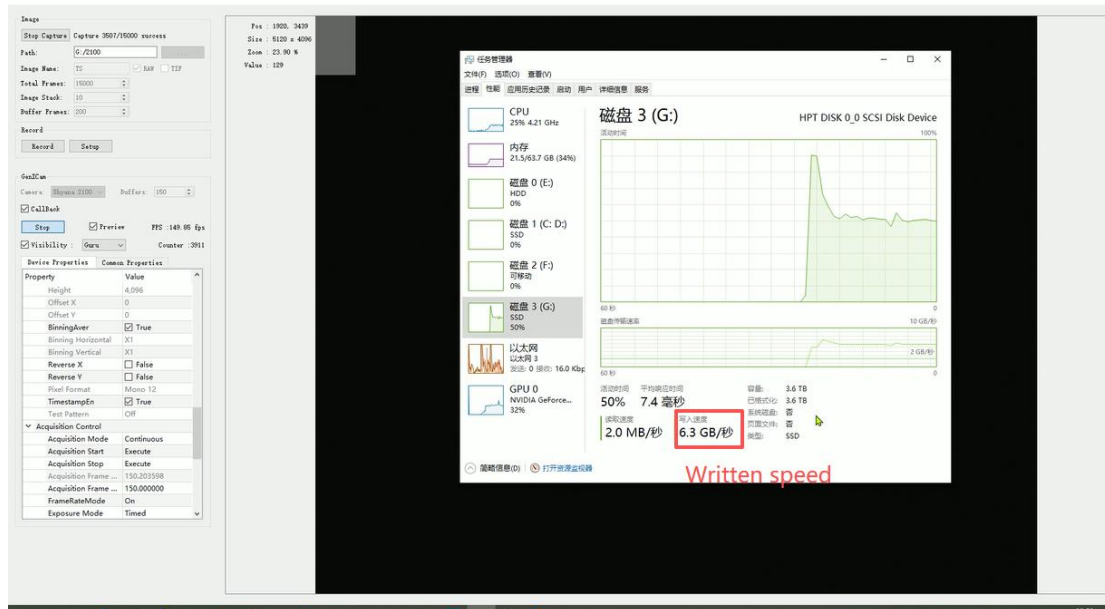
Buffer Frames: 200 [one unit means one frame]

Buffer: 150 [one unit means one frame]

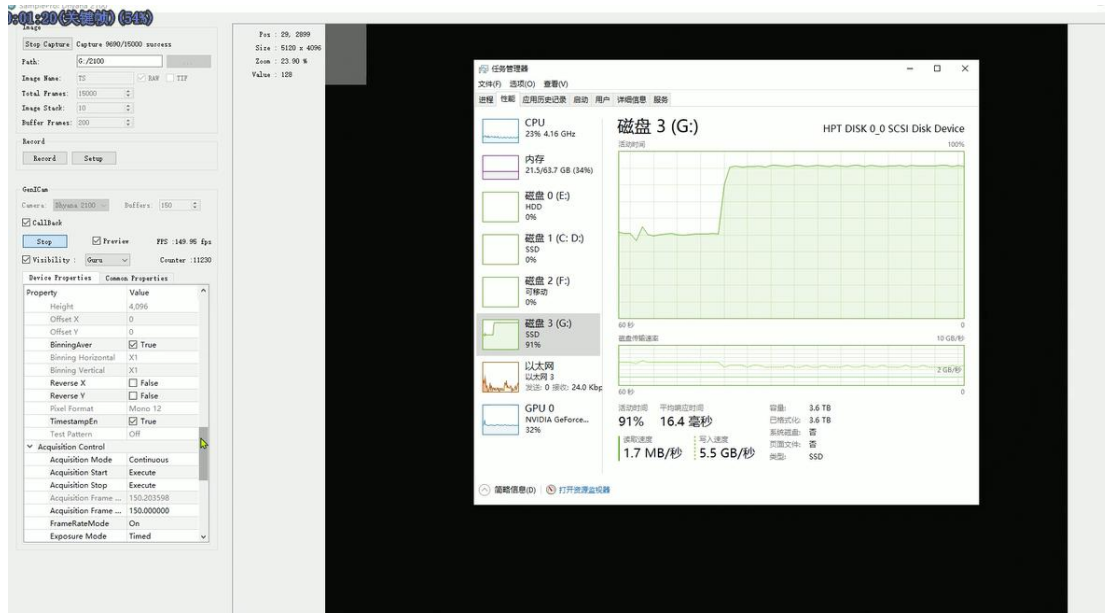
Trigger Mode:OFF

3. Test result (Live mode, trigger off)

A camera with a resolution of 4096×5120 , 16-bit image storage depth, and 150 fps generates a data rate of approximately 6.29 GB/s.

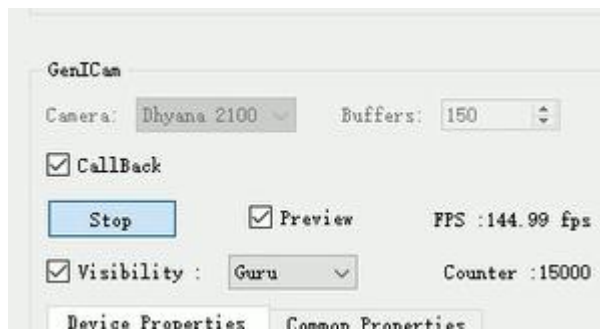


The write speed can reach the highest 6.3GB/S.



After writing for a while, we do see the write speed drop between 5 and 5.5GB due to the temperature degradation. In our PC, we didn't observe that the write speed is limited at 4GB. In the next period, we will figure out your configuration, and see if there is any mismatched parameters settings.

And every trigger is collected by the camera.

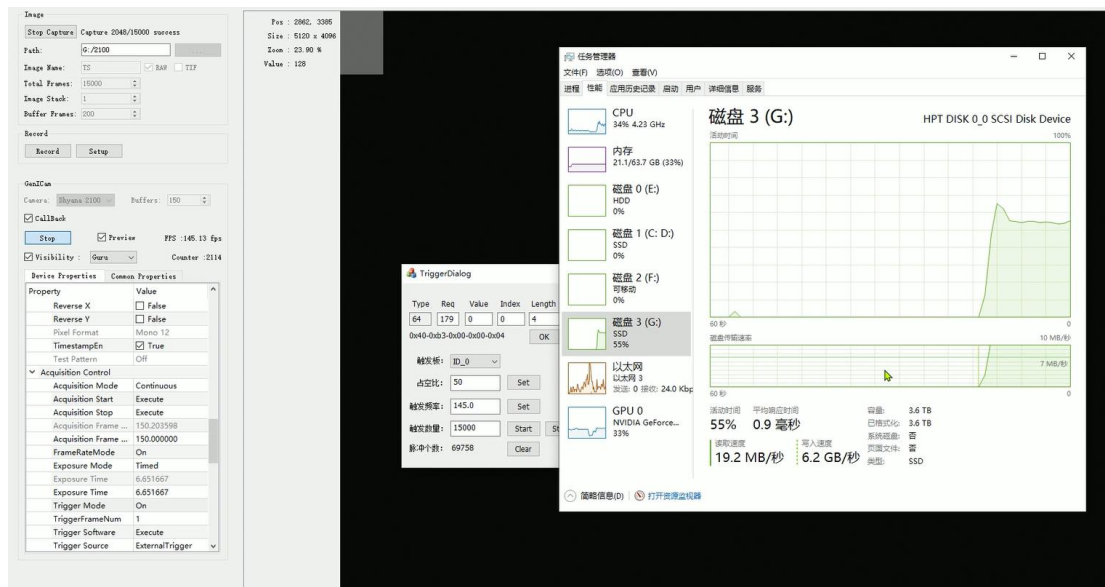


4. Test result (Turn off Live mode, trigger on)

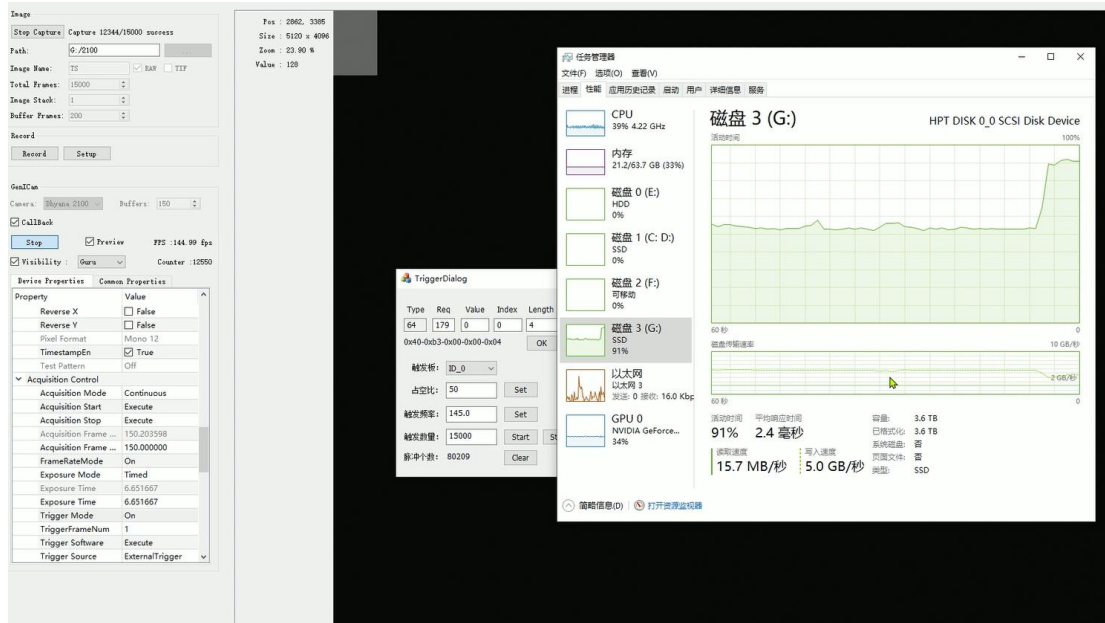
For improving the frame drop in the live mode, in this part, we use the trigger to capture the frame.

Trigger Frequency: 145HZ

External Trigger Number: 150000



Initially, the write speed is nearly 6.2 GB/s, which is close to the theoretical transfer rate. However, after approximately 12,000 frames are captured, the speed drops due to the heat accumulation in the RAID array.



5. Verdict

In our win system, we didn't observe the camera have strongly frame loss in storing the images into the Raid array before the Raid array is too hot. In the next period, we will confirm the Raid Array setups with you, we will coordinate with you to solve your problem.